NETSOL 2.1 Date________ M T W T F S S

1)(a) $\forall x[\exists y F(x,y) \rightarrow S(x)]$ Pg 66

(b) ~(anybody in calculus class is smarter than ~~everyone~~ anyone in DS class)

$\sim \exists x[C(x) \wedge \forall y(D(y) \rightarrow S(x,y))]$

(c) $\forall x[\sim(x=M) \rightarrow L(x,M)]$

(d) $\exists x[P(x) \wedge S(j,x)] \wedge \exists y[P(y) \wedge S(r,y)]$

(e) $\exists x[P(x) \wedge S(j,x) \wedge S(r,x)]$

2)(a) ∀x(if x bought Rolls Royce with cash x has rich uncle)

$\forall x[R(x) \rightarrow U(x)]$. → or $\forall x[(R(x) \wedge C(x)) \rightarrow U(x)]$

with cash.

(b) $\exists x(D(x) \wedge M(x)) \rightarrow \forall y[\exists x(F(y,x) \wedge D(x)) \rightarrow Q(y)]$

(c) $\sim \exists x F(x) \rightarrow \forall y[A(y) \rightarrow \exists x\, T(y,x)]$

(d) means nobody can do it except Jones

$\sim \exists x[D(x) \wedge \sim J(x)]$

It is not the case that there exists any x such that x can do it & x is not Jones.

(e) $\exists x[J(x) \wedge D(x)] \rightarrow \forall x\, D(x)$

3)(a) $\forall z[\exists x L(x,x) \rightarrow \exists y L(z,y)]$

z is bound, x & y are free

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(b) $\forall a [\exists x \; ax^2 + 4x - 2 = 0 \longleftrightarrow a \geq -2]$ x is free

(c) $\forall x [x^3 - 3x < 3 \rightarrow x < 10]$ No free

Note that this automatically means that x is solution (similar with part (b))

(d) $\forall w \; (\exists x \; x^2 + 5x = w \wedge \exists y \; 4 - y^2 = w) \rightarrow (1 \leq w < 10)$

w is free (To know if this statement is T, we need to know what w is, not x & y)

Q4)

(a) Every man who is unmarried is unhappy. or All unmarried men are unhappy.

(b) y is a sister of one of x's parents.

5) (a) Every prime number except 2 is odd.

(b) There is a perfect number x such that all perfect numbers except the x are less than x.

or x is a perfect number & all perfect numbers are less than or equal to x

6) (a) There is no number x that satisfies both $x^2 + 2x + 3 = 0$ and $x^2 + 2x - 3 = 0$ False

(b) It is not the case that there is a value of x that satisfies $x^2 + 2x + 3$ and there is a value of x that satisfies $x^2 + 2x - 3 = 0$

more good: It is not the case that both $x^2 + 2x + 3 = 0$ & $x^2 + 2x - 3 = 0$ has real solutions.

or Atleast one of eq $x^2 + 2x + 3$ & $x^2 + 2x - 3$ has no solution

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As one of the eq has no real solution while other has the true.

(c) There is no x that satisfies $x^2+2x+3=0$ and there is no x that satisfies $x^2+2x-3=0$

OR $x^2+2x+3=0$ and x^2+2x-3 both have no real solutions is false

4)(a) On Book

3. Analyze the logical forms of the following statements. The universe of discourse is $\mathbb{R}$. What are the free variables in each statement?
   (a) Every number that is larger than x is larger than y.
   (b) For every number a, the equation $ax^2 + 4x - 2 = 0$ has at least one solution iff $a \geq -2$.
   (c) All solutions of the inequality $x^3 - 3x < 3$ are smaller than 10.
   (d) If there is a number x such that $x^2 + 5x = w$ and there is a number y such that $4 - y^2 = w$, then w is strictly between -10 and 10.

*4. Translate the following statements into idiomatic English.
   (a) $\forall x[(H(x) \wedge \neg\exists y M(x, y)) \rightarrow U(x)]$, where $H(x)$ means "x is a man," $M(x, y)$ means "x is married to y," and $U(x)$ means "x is unhappy."
   (b) $\exists z(P(z, x) \wedge S(z, y) \wedge W(y))$, where $P(z, x)$ means "z is a parent of x," $S(z, y)$ means "z and y are siblings," and $W(y)$ means "y is a woman."

5. Translate the following statements into idiomatic mathematical English.
   (a) $\forall x[(P(x) \wedge \neg(x = 2)) \rightarrow O(x)]$, where $P(x)$ means "x is a prime number" and $O(x)$ means "x is odd."
   (b) $\exists x[P(x) \wedge \forall y(P(y) \rightarrow y \leq x)]$, where $P(x)$ means "x is a perfect number."

6. Translate the following statements into idiomatic mathematical English. Are they true or false? The universe of discourse is $\mathbb{R}$.
   (a) $\neg\exists x(x^2 + 2x + 3 = 0 \wedge x^2 + 2x - 3 = 0)$.
   (b) $\neg[\exists x(x^2 + 2x + 3 = 0) \wedge \exists x(x^2 + 2x - 3 = 0)]$.
   (c) $\neg\exists x(x^2 + 2x + 3 = 0) \wedge \neg\exists x(x^2 + 2x - 3 = 0)$.

7. Are these statements true or false? The universe of discourse is the set of all people, and $P(x, y)$ means "x is a parent of y."
   (a) $\exists x \forall y P(x, y)$. F
   (b) $\forall x \exists y P(x, y)$. F
   (c) $\neg\exists x \exists y P(x, y)$. F (There is no child without parent)
   (d) $\exists x \neg\exists y P(x, y)$. T (some parents dont have children)
   (e) $\exists x \exists y \neg P(x, y)$. T

*8. Are these statements true or false? The universe of discourse is $\mathbb{N}$.
   (a) $\forall x \exists y(2x - y = 0)$. T
   (b) $\exists y \forall x(2x - y = 0)$. F → it says there exists some y such that for all x, this eq holds
   (c) $\forall x \exists y(x - 2y = 0)$. F (e.g. x=1 → y is Natural values)
   (d) $\forall x(x < 10 \rightarrow \forall y(y < x \rightarrow y < 9))$. T
   (e) $\exists y \exists z(y + z = 100)$. T
   (f) $\forall x \exists y(y > x \wedge \exists z(y + z = 100))$. F (x can grow very large exceeding 100)

9. Same as exercise 8 but with $\mathbb{R}$ as the universe of discourse. T, F, T, T, T, T (for -ve z)

10. Same as exercise 8 but with $\mathbb{Z}$ as the universe of discourse. T, F, F, T, T, T